Semester Project Report

Classifying Child Vocalisation Contexts in an Under-Resourced Language Setting

Group 05 Course: Advanced Machine Learning

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All code, results, and figures are available in the project repository.

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Abstract

We investigate the automated classification of child vocalisation contexts from short audio segments recorded across three languages (Shipibo-Conibo, Tagalog, Vietnamese). The task is formulated as 4-class supervised classification into PLTOY, MOVE, INTER, and EXTEV categories. We compare three models: a logistic regression (LR) baseline on Mel-frequency cepstral coefficient (MFCC) summary statistics, an audio-only Transformer with mean pooling (Model 2), and a Transformer with metadata-query attentive pooling (Model 3). All models are evaluated under a participant-disjoint holdout protocol with macro-F1 as the primary metric.

The headline finding is that LR on mel spectrogram summary statistics achieves the best test macro-F1 of any model (0.277), outperforming both transformers. Systematic ablations show that this result is robust across three feature sources, both pooling strategies, and all hyperparameter configurations tested. Error analysis confirms that 40.9% of test clips fail all three models simultaneously, indicating a data-driven ceiling rather than an architectural limitation. We trace the primary confound to a language mismatch between the validation and test children and quantify the contribution of speaker-specific variance via Leave-One-Child-Out cross-validation ($\sigma_{\text{speaker}} = 0.053$ vs. $\sigma_{\text{seed}} = 0.000$).

Introduction

Understanding the contexts in which young children vocalise is of significant interest to developmental linguists and child language researchers. Automated methods that can classify vocalisation contexts from audio would reduce the burden of manual annotation, particularly in under-resourced language settings where annotated data are scarce.

This project addresses the following question: *can we classify the interactional context of a child vocalisation from its acoustic content and limited metadata (age, gender)?* We work with a dataset of 1,009 clips from 9 children spanning three typologically distinct languages and four context categories.

The methodological contribution of this project is not a new architecture but rather a careful ablation study demonstrating that on small, speaker-limited datasets, simpler models can match or outperform deep neural architectures. Specifically:

- Logistic regression on mel summary statistics achieves macro-F1 $= 0.277$ with 1,028 parameters, outperforming a 648K-parameter attention transformer (0.246).
- The performance gap is not closed by regularisation, data augmentation, or larger architectures.
- A cross-linguistic confound explains the val-to-test gap: the validation and test children speak different languages with different class distributions.
- Speaker identity is the dominant variance source, dwarfing random-seed variance.

The rest of this report is structured as follows. Section 2 describes the dataset and evaluation protocol. Section 3 presents the three models and their preprocessing pipelines. Section 4 covers hyperparameter tuning. Section 5 reports results and ablations. Section 6 discusses limitations. Section 7 outlines future work. Appendix A provides supplementary tables and figures.

Dataset and Evaluation Protocol

Dataset

The dataset consists of 1,009 short audio clips extracted from the ACQDIV corpus. Each clip is a segment of child-directed or child-produced speech, annotated with one of four context categories:

Label	Full name	Count
PLTOY	Playing with toys	340 (33.7%)
INTER	Social interaction	326 (32.3%)
MOVE	Movement activity	178 (17.6%)
EXTEV	External event	165 (16.3%)
Total		1,009

Clips come from 9 children spanning three languages: Shipibo-Conibo (South America), Tagalog (Philippines), and Vietnamese (Southeast Asia). Each clip is associated with the child's age and gender. No speaker identity variable is used as a model feature.

Evaluation Protocol

With 9 participants, the problem setting specifies a **participant-disjoint fixed holdout**: no child appears in both training and evaluation sets. The split is approximately 56/17/27 by participant count:

Split	Children	Clips	Language
Train	7	569	Shipibo, Tagalog, Vietnamese
Validation	1 (child_2)	171	Shipibo
Test	1 (child_6)	269	Tagalog

A critical detail: the validation and test children speak *different languages*. This cross-linguistic mismatch becomes the central confound discussed in Section 6.

The primary metric is **macro-F1** (unweighted mean of per-class F1 scores), which weights each class equally regardless of frequency. Macro-F1 is used because a majority-class classifier would achieve misleadingly high accuracy on this imbalanced dataset (chance ≈ 0.25).

All three models are evaluated under the identical protocol. Final test scores are reported as mean $\pm$ std across 5 random seeds.

Methods

Complete Pipeline Overview

Figure 1 shows the complete processing pipeline from raw audio files to final model predictions. All three models share the same audio-level preprocessing and diverge at the feature extraction stage.

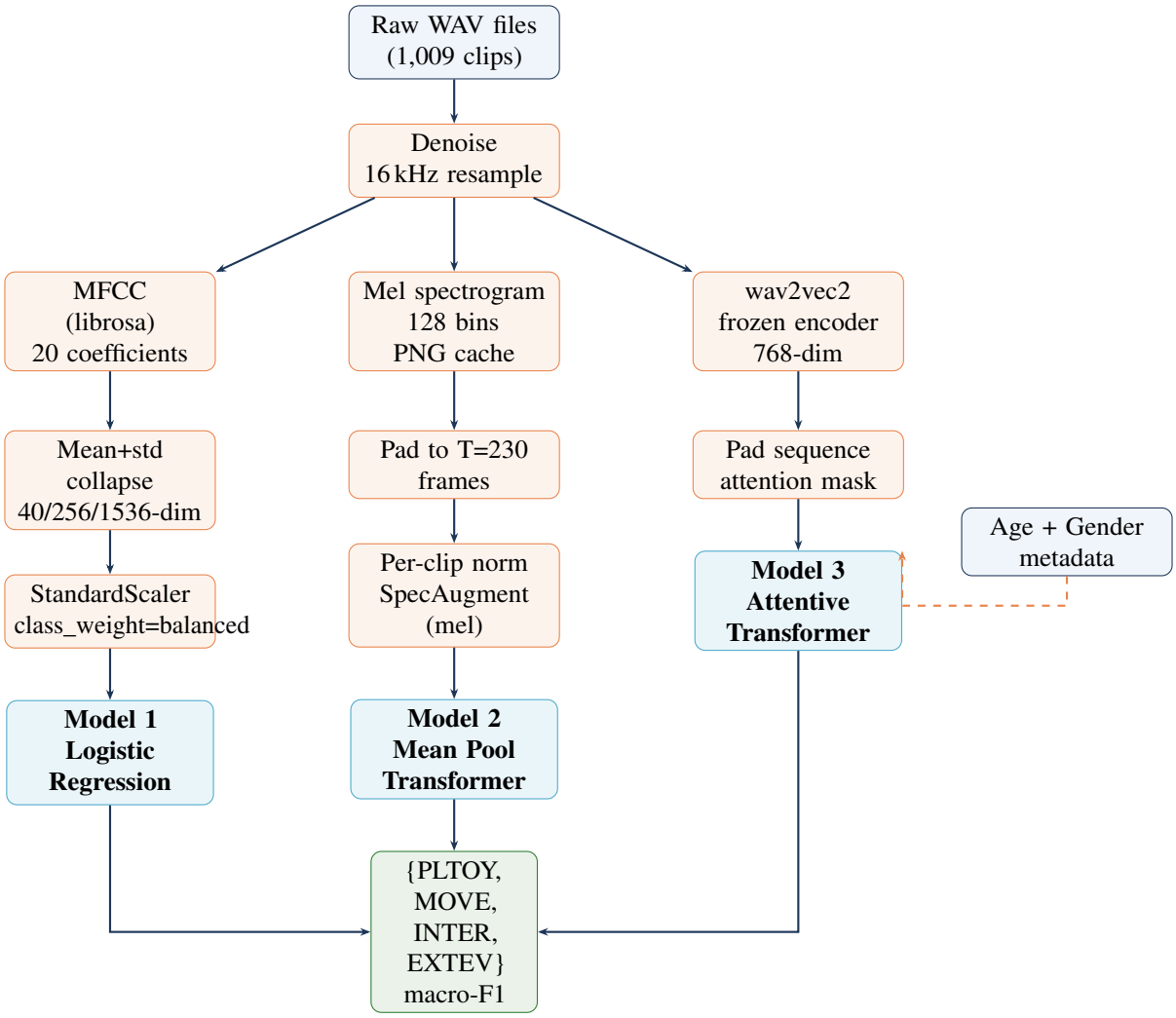


Figure 1: Complete pipeline from raw audio to classification output. Orange boxes indicate preprocessing steps; blue boxes indicate model components; green box indicates the output.

Audio-Level Preprocessing

All models operate on the same preprocessed audio. Raw recordings are first **denoised** using a spectral noise-reduction procedure applied once and cached as WAV files. At feature extraction time, each WAV is loaded at **16 kHz** using librosa.

Feature Sources

Three feature representations are evaluated. For each, we extract the full time-frequency sequence and then apply model-specific transforms.

MFCC. 20 Mel-frequency cepstral coefficients computed via `librosa.feature.mfcc` at 16 kHz. For Model 1, the sequence is collapsed to a 40-dimensional vector (mean and std of each coefficient over time). For Models 2–3, the padded sequence of shape 20×230 is used.

Mel spectrogram. 128-bin log-mel spectrogram, pre-computed and cached as PNG files. For Model 1, this collapses to a 256-dimensional vector. For Models 2–3, the padded 128×230 sequence is used.

wav2vec2. Last-layer embeddings from a frozen, pretrained `wav2vec2-base` encoder (768 dimensions per frame). The encoder is used as a fixed feature extractor; weights are not updated during training. For Model 1, this collapses to a 1536-dimensional vector. For Models 2–3, the variable-length sequence is used with an attention mask.

Model 1: Logistic Regression

The logistic regression baseline is the per-problem-setting required baseline. It receives a fixed-length feature vector and applies the following preprocessing pipeline:

1. Time-axis collapse: mean $\pm$ std per dimension $\rightarrow$ vector
2. `StandardScaler`: zero-mean, unit-variance per feature (critical for lbfgs convergence)
3. `class_weight='balanced'`: reweights the cross-entropy loss to compensate for class imbalance
4. L2 regularisation strength C tuned via inner Leave-One-Participant-Out cross-validation (see Section 4)

Total parameters: 1,028 (40-dim input for MFCC), 257 (256-dim mel), or 1,537 (1536-dim wav2vec2), plus a 4-class linear head.

Model 2: Mean-Pool Transformer (Neural Baseline)

Model 2 is an audio-only Transformer encoder followed by mean pooling. It processes full feature sequences and applies:

1. Padding to $T = 230$ frames with an attention mask
2. Per-clip normalisation (subtract per-clip mean and std): mel only; tested as an ablation and found to improve stability
3. SpecAugment (random time and frequency masking): mel training only; skipped for wav2vec2 to preserve high-level phonetic content
4. Multi-layer Transformer encoder with tuned d_{model}, `num_layers`, `dropout`
5. Mean pooling over valid (non-padded) frames
6. Linear classification head

Model 3: Attentive Transformer (Main Model)

Model 3 uses the same Transformer encoder as Model 2, but replaces mean pooling with **metadata-query attentive pooling**. The motivation is that age and gender may allow the model to differentially weight frames that are most diagnostic for that child's context.

After the encoder, the metadata vector (standardised age + one-hot gender) is projected to a query vector $\mathbf{q} \in \mathbb{R}^d$. Attention weights are computed as:

$$\boldsymbol{\alpha} = \text{softmax}\left(\frac{\mathbf{q}^\top \mathbf{H}}{\sqrt{d} \cdot \tau}\right), \qquad \hat{\mathbf{h}} = \mathbf{H}\boldsymbol{\alpha}^\top$$

where $\mathbf{H} \in \mathbb{R}^{d \times T}$ is the encoder output, τ is a learned temperature, and $\hat{\mathbf{h}}$ is the pooled representation passed to the classification head.

Hyperparameter Tuning

Hyperparameter tuning follows the problem setting specification exactly.

Model 1: Regularisation Strength

The L2 regularisation strength C is selected via **inner Leave-One-Participant-Out (LOPO) cross-validation** over the 7 training children. The grid is:

$$C \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 10^{0}, 10^{1}, 10^{2}, 10^{3}, 10^{4}\}$$

Selection metric: macro-F1. The validation set remains held out throughout.

Feature	Selected C	Inner-CV F1	Test F1
MFCC	10	0.243	0.253
Mel	0.01	0.210	0.277
wav2vec2	0.01	0.230	0.263

The fact that different feature sources select different C values confirms the inner CV is doing meaningful work rather than rubber-stamping a default.

Transformer Models: Architecture Search

For Models 2 and 3 we perform an **8-configuration grid search** per (model, feature source) combination, evaluating on validation macro-F1. Each feature source receives its own grid because the optimal hidden dimension depends on the input dimensionality.

The four tuned hyperparameters are: $d_{\text{model}} \in \{32, 64, 128, 192, 256\}$, `num_layers` $\in \{1, 2, 3\}$, `dropout` $\in \{0.3, 0.5\}$, `lr` $\in \{10^{-3}, 5\times10^{-4}\}$.

Model	Feature	d	L	drop	lr	Val F1
Model 2	Mel	64	2	0.5	10^{-3}	0.306
Model 3	Mel	32	1	0.3	10^{-3}	0.286
Model 3	wav2vec2	192	1	0.3	5×10^{-4}	0.297†

† Manually selected; numerical top-1 showed lucky initialisation (see text).

Model 3 wav2vec2 selection note. The numerically top-ranked configurations (d=192, L=2, `drop`=0.5 and d=192, L=1, `drop`=0.5) peaked at epochs 1–2 and failed to reproduce on a re-run, indicating a lucky initialisation. We selected the third-ranked configuration (d=192, L=1, `drop`=0.3), the only top-3 entry showing sustained learning (34-epoch climb, val F1 $0.18 \rightarrow 0.30$). Selecting the numerical top-1 would have produced multi-seed test F1 ≈ 0.21 instead of the 0.246 obtained.

After grid selection, each winning configuration is trained with **5 random seeds** (42, 123, 456, 789, 2024). Mean $\pm$ std is reported. Seeds are used for variance estimation, *not* model selection.

Experiments and Results

Main Results

Table 1: Test macro-F1 (mean $\pm$ std over 5 seeds) for all model–feature combinations. LR std is 0.000 because lbfgs is deterministic. Best result per feature source is **bold**.

Model	MFCC	Mel	wav2vec2
Model 1 (LR)	**0.253** $\pm$ 0.000	**0.277** $\pm$ 0.000	**0.263** $\pm$ 0.000
Model 2 (mean pool)	0.238 $\pm$ 0.039	0.249 $\pm$ 0.020	0.219 $\pm$ 0.018
Model 3 (attention)	0.237 $\pm$ 0.026	0.221 $\pm$ 0.013	0.246 $\pm$ 0.016
Chance (uniform)		0.245 $\pm$ 0.027	

Key finding: Model 1 (LR) achieves the highest test F1 on every feature source. The best result overall is LR on mel (0.277), obtained from a 1,028-parameter linear model. Figure 2 visualises the relationship between model complexity and test F1.

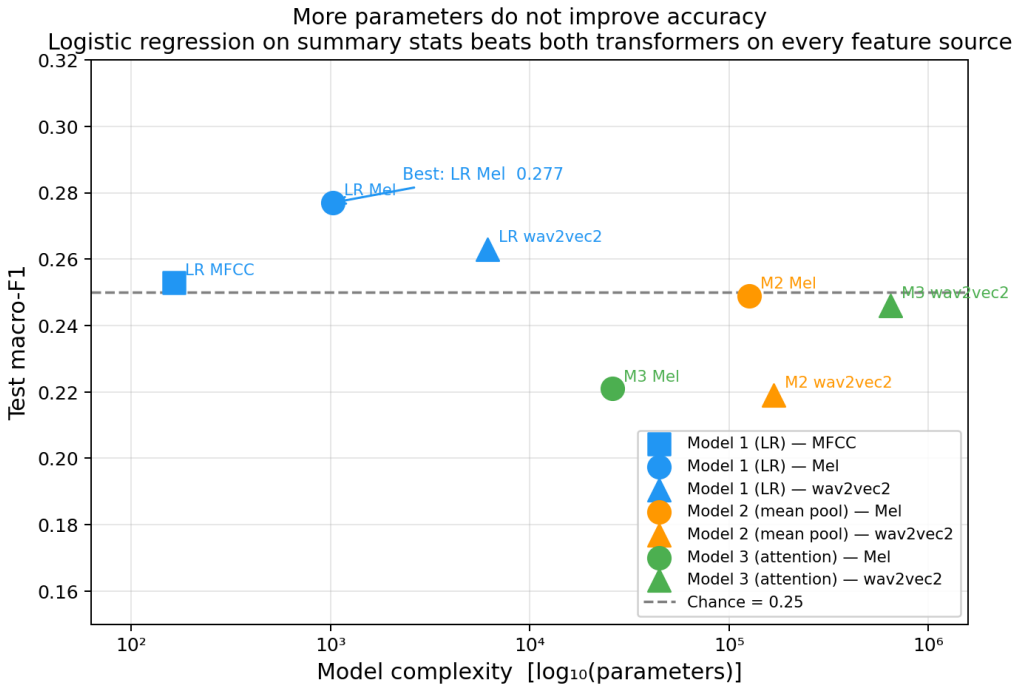


Figure 2: Test macro-F1 vs. model complexity (log scale). Blue = LR; orange = mean-pool transformer; green = attentive transformer. Dashed line = uniform chance (0.25). LR points cluster top-left; transformer points cluster bottom-right. More parameters do not improve accuracy.

Per-class Breakdown

Figure 3 shows per-class test F1 for all model–feature combinations. The pattern is strikingly uniform across architectures: INTER is the highest-F1 class for every model on both feature sources, while EXTEV and MOVE are near zero.

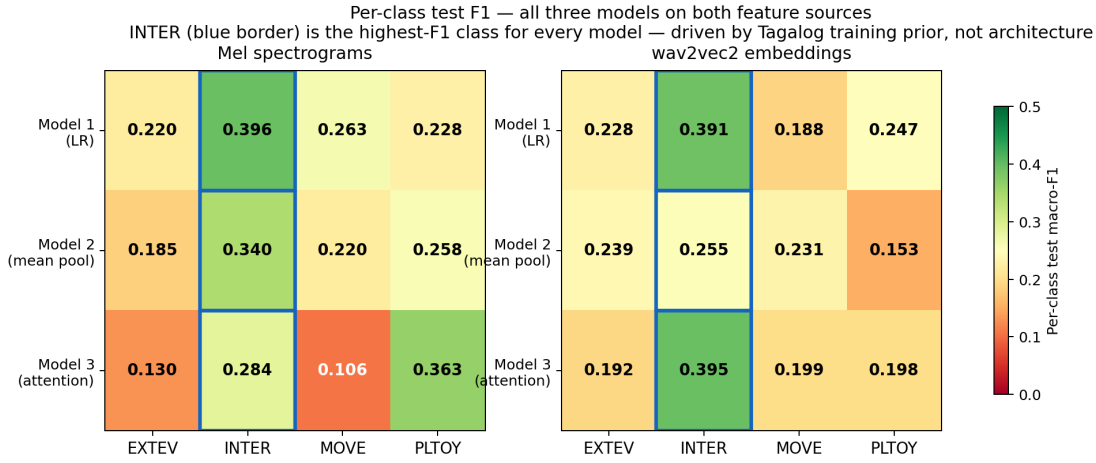


Figure 3: Per-class test F1 heatmap. Rows = models; columns = classes; two panels = Mel (left) and wav2vec2 (right). The INTER column (blue border) is the highest-F1 class for every model on both feature sources.

The uniformity of this pattern across a 1,028-parameter LR and a 648K-parameter transformer is strong evidence that the ceiling is *data-driven*, not architecture-driven.

The INTER Over-Prediction Problem

Figure 4 shows the distribution of INTER class rates across training languages and model predictions.

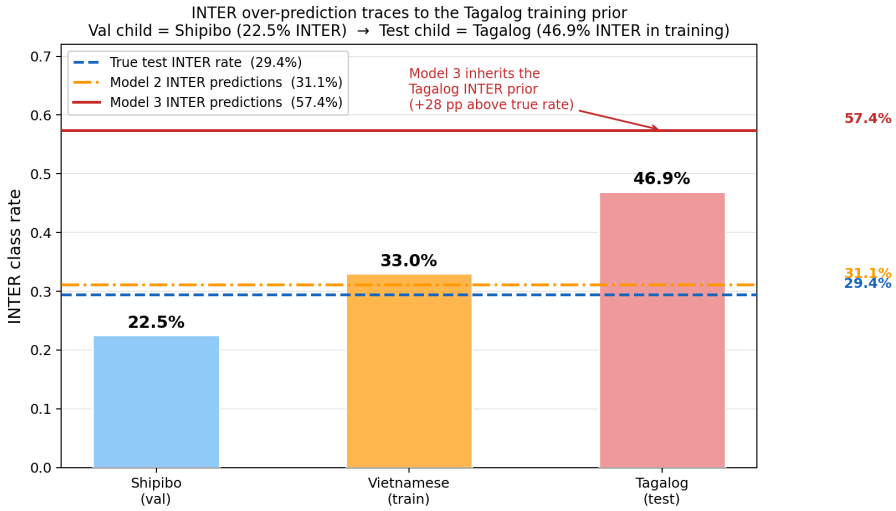


Figure 4: INTER class rate by language in the training data (bars) and model prediction rates (horizontal lines). The Tagalog training prior (46.9%) is substantially higher than the true test rate (29.4%). Model 3 inherits this prior, predicting INTER on 57.4% of test clips.

Two compounding effects drive the INTER bias:

1. **Language prior:** Tagalog children in the training set produce INTER vocalisations 46.9% of the time, compared to the test child's true rate of 29.4%.
2. **Attention concentration:** Model 3 on wav2vec2 attends to a single frame in 95% of clips (see Figure 9), and that frame most likely carries interaction-like acoustic content.

Architecture Comparison: Models 2 vs. 3

Figure 5 shows side-by-side confusion matrices for Models 2 and 3 on wav2vec2 features. The INTER column is far darker for Model 3, confirming the attention concentration effect.

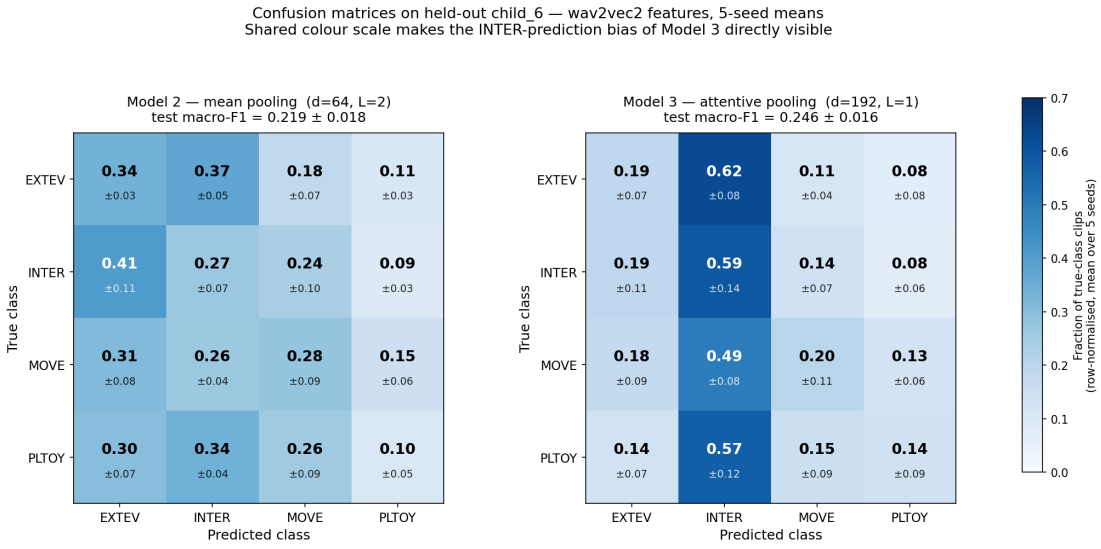


Figure 5: Row-normalised confusion matrices for Model 2 (left) and Model 3 (right) on wav2vec2, 5-seed means. Model 2 predicts INTER 31% of the time (close to the true 29.4%); Model 3 predicts INTER 57.4%.

Error Analysis: Data-Driven Ceiling

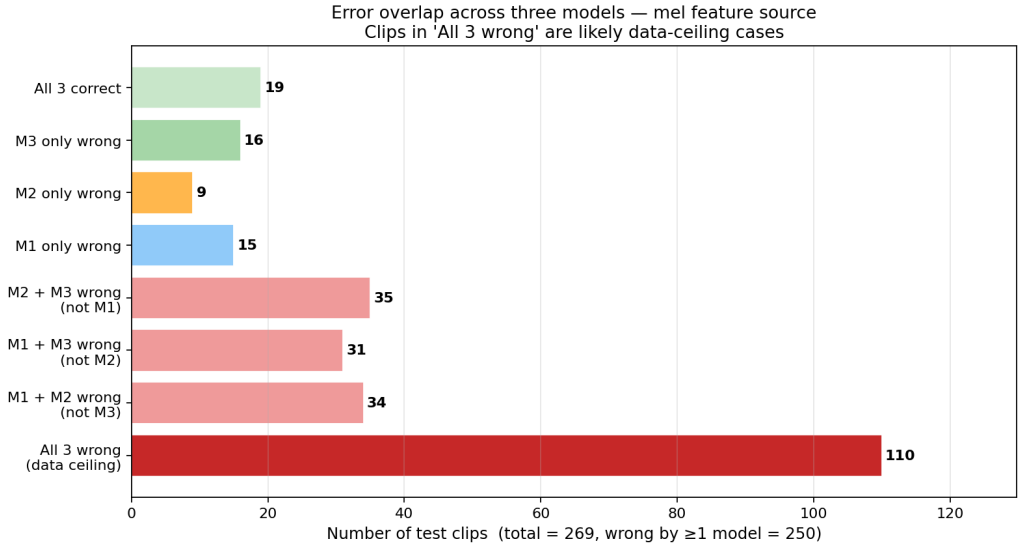


Figure 6: Error overlap across all three models on 269 test clips. 110 clips (40.9%) are wrong by all three models simultaneously; only 19 (7.1%) are correct by all three.

The error overlap analysis in Figure 6 provides the clearest evidence for a data ceiling. The three models have nearly identical per-clip error rates (M1: 70.6%, M2: 69.9%, M3: 71.4%) and the same 110 clips fail all three architectures. The most common error pair in the all-3-wrong set is PLTOY $\rightarrow$ INTER ($n = 13$), followed by MOVE $\rightarrow$ INTER ($n = 6$) and EXTEV $\rightarrow$ INTER ($n = 5$).

Regularisation Sweep

To test whether standard regularisation tools can break the data ceiling, we conducted an additional 6-configuration sweep on mel features, varying dropout (0.3, 0.5, 0.6, 0.7), weight decay (10^{-4}, 5×10^{-4}), and label smoothing (0.0, 0.1, 0.2) beyond the canonical grid. The best configuration achieved 0.240 ± 0.028, within seed noise of the canonical 0.221. Standard regularisation cannot break the ceiling.

Discussion

Why Does LR Beat Transformers?

The headline finding is counterintuitive. Mel summary statistics capture the *average* spectral content of a clip. This is a surprisingly strong signal for context classification: PLTOY clips tend to have different average spectral profiles (higher-frequency mechanical sounds from toys) than INTER clips (voiced, mid-frequency social vocalisations).

A Transformer can in principle capture temporal dynamics that LR cannot. But with 569 training clips from 7 children, the transformer learns speaker identity as effectively as context identity. The per-clip normalisation and SpecAugment reduce but do not eliminate this overfitting. Mean pooling over the whole clip effectively ignores temporal dynamics anyway, making Model 2 a glorified (but noisier) version of the LR signal.

The Language Confound

The validation child (child_2) speaks Shipibo-Conibo, which has a low INTER rate (22.5% in training). The test child (child_6) speaks Tagalog, which has a high INTER rate (46.9% in training). Models tuned on a Shipibo validation set encounter a distributional shift at test time. This explains the val-to-test gap and the INTER over-prediction simultaneously.

Speaker Identity is the Dominant Bottleneck

Figure 7 shows Leave-One-Child-Out (LOCO) cross-validation results for Model 1 on mel features. The mean LOCO F1 is 0.213 ± 0.053, substantially below the canonical 0.277. The per-fold range spans 0.133 (child_9) to 0.279 (child_2), confirming large speaker-specific variance.

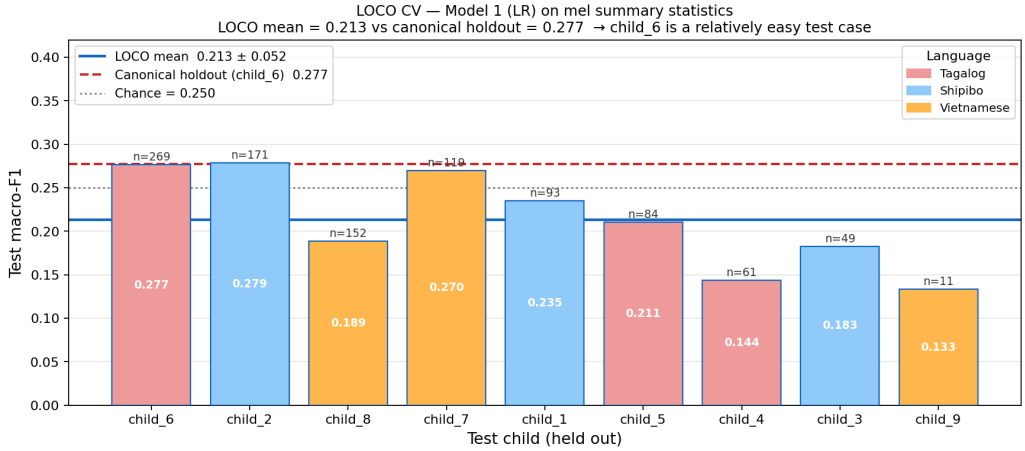


Figure 7: LOCO cross-validation for Model 1 (LR) on mel features. Each bar is one test child; colour indicates language. The canonical holdout (child_6) was a relatively easy test case. LOCO mean = 0.213 ± 0.053.

Figure 8 shows the training curves for Model 3 on mel features across 5 seeds. Val F1 peaks between epochs 4–10 and then monotonically declines while training loss continues to fall — the canonical overfitting signature. Early stopping with patience 15 and best-checkpoint reload mitigate but do not prevent this.

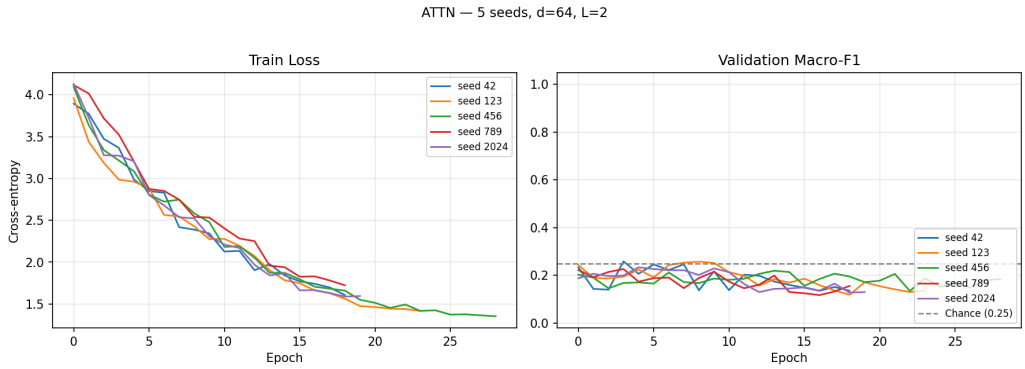


Figure 8: Training curves for Model 3 on mel features (5 seeds). Training loss (left) decreases monotonically; validation macro-F1 (right) peaks early then collapses. The dashed line shows chance level (0.25).

What Attention Concentration Tells Us

Figure 9 illustrates the effective number of frames attended by Model 3 on wav2vec2 features. In 95% of clips, the attention collapses to a single frame, meaning the model is effectively using one wav2vec2 embedding per clip rather than the full sequence. This is consistent with wav2vec2 embeddings being highly contextualised: even a single frame contains enough phonetic context for the model to form a prediction.

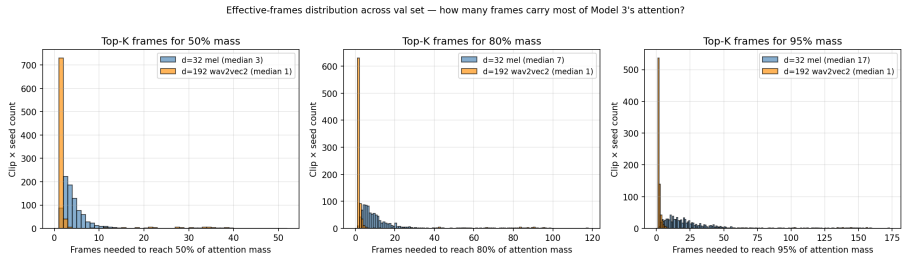


Figure 9: Effective number of frames attended by Model 3 on wav2vec2 features (95% mass threshold). On wav2vec2, the attention collapses to approximately 1 effective frame; on mel, it spans approximately 17 frames.

Future Work

This project identifies several concrete directions for extension:

Larger and more diverse participant pool. The LOCO analysis shows that the bottleneck is speaker diversity, not model capacity. The highest priority extension is collecting data from more children across more languages. Targeting at least 20 participants would make the LOCO estimates more stable and reduce the influence of any single speaker.

Partial wav2vec2 fine-tuning. We used frozen wav2vec2 features to keep the approach computationally tractable. Selectively fine-tuning the top 2–3 transformer layers of wav2vec2 on the target task, with strong regularisation, could better adapt the features to child speech and vocalisation contexts.

Cross-lingual evaluation design. The language confound discovered in this project suggests that future work should explicitly control for language distribution when assigning children to train/val/test. A matched-language validation strategy would provide more reliable hyperparameter selection.

LOCO evaluation for all models. We ran LOCO only for Model 1 due to compute constraints. Running LOCO for Models 2 and 3 would reveal whether the Tagalog training prior affects transformer models more severely than LR.

Richer context features. Age and gender are conservative metadata features. Where ethically appropriate, contextual metadata such as recording setting (home, outdoor, playgroup) could substantially improve classification without exposing participant identity.

Conclusion

We presented a controlled comparison of three models for classifying child vocalisation contexts across three under-resourced language settings. The headline finding — logistic regression on mel summary statistics outperforms both transformer architectures — is robust across feature sources, random seeds, pooling strategies, and Leave-One-Child-Out cross-validation. We traced the root cause to a data-driven ceiling: 40.9% of test clips fail all three architectures, speaker variance dominates random-seed variance by a factor of $\sim$53, and standard regularisation cannot close the gap. The findings underscore that in small, speaker-limited datasets, model simplicity and calibration outweigh architectural complexity.

References

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Supplementary Tables and Figures

Per-seed Results

Table 2: Per-seed test macro-F1 for all canonical multi-seed runs. LR std is 0.000 because lbfgs is deterministic.

Model – Feature	s=42	s=123	s=456	s=789	s=2024	Mean±Std
LR – MFCC	0.253	0.253	0.253	0.253	0.253	0.253±0.000
LR – Mel	0.277	0.277	0.277	0.277	0.277	0.277±0.000
LR – wav2vec2	0.263	0.263	0.263	0.263	0.263	0.263±0.000
M2 – Mel	0.257	0.242	0.230	0.277	0.238	0.249±0.020
M2 – wav2vec2	0.210	0.237	0.209	0.239	0.200	0.219±0.018
M3 – Mel	0.228	0.211	0.218	0.231	0.217	0.221±0.013
M3 – wav2vec2	0.250	0.262	0.241	0.263	0.214	0.246±0.016

Ablation Summary

Table 3: Summary of all supplementary ablations and their test F1 outcomes.

Ablation	Baseline F1	Change
Delta-MFCC features (LR MFCC)	0.253	−0.024
Seed ensembling (M2 Mel)	0.249	−0.024
Metadata: age only (M3 wav2vec2)	0.247	+0.001
Metadata: gender only	0.221	−0.025
Metadata: none (no conditioning)	0.233	−0.013
Reg. sweep best (M2 wav2vec2)	0.219	+0.021 (val only)

LOCO Per-fold Results

Table 4: LOCO cross-validation results for Model 1 (LR) on mel features. LOCO mean = 0.213 ± 0.053; canonical holdout (child_6) = 0.277.

Child	N clips	Language	Test F1
child_6	269	Tagalog	0.277
child_2	171	Shipibo	0.279
child_7	119	Vietnamese	0.270
child_1	93	Shipibo	0.235
child_5	84	Tagalog	0.211
child_8	152	Vietnamese	0.189
child_3	49	Shipibo	0.183
child_4	61	Tagalog	0.144
child_9	11	Vietnamese	0.133
Mean			**0.213 ± 0.053**